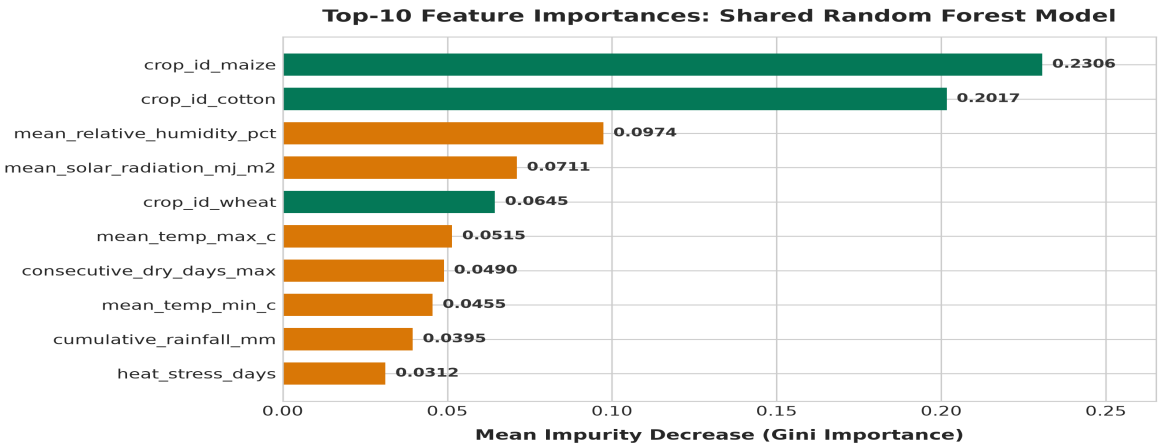
Agricultural Crop Yield Prediction: Model Analysis & Evaluation

Universal Multi-Crop Random Forest Model • Maharashtra Agroclimatology & ISRIC SoilGrids 2.0 • Presentation Report

1. Feature Importance: Top-10 Features with Scores (Random Forest Gini)

Feature importances extracted from the trained **Shared Multi-Crop Random Forest Regressor**. Scores represent mean decrease in impurity (Gini importance) across all 100 decision trees.

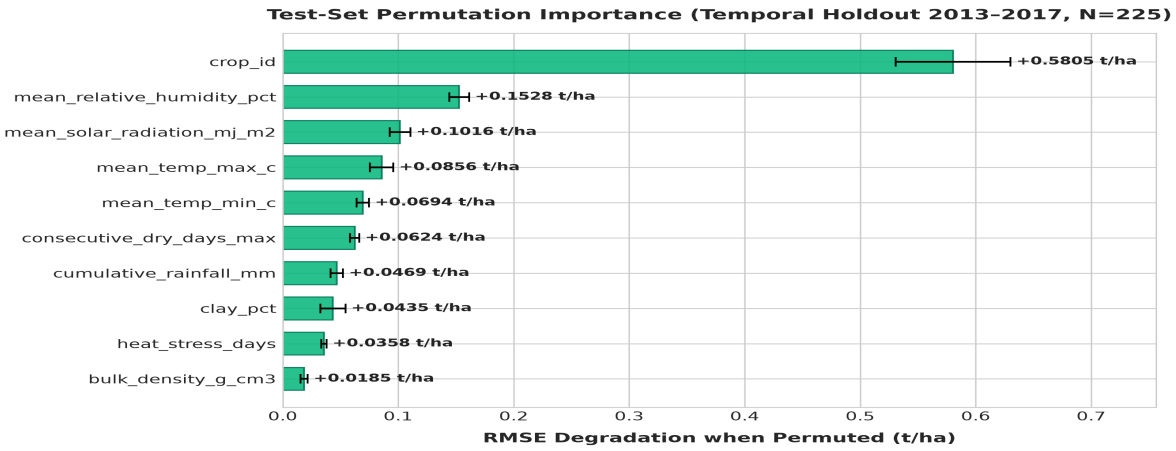
Rank	Feature Name	Gini Importance Score	Category / Domain
1	**crop_id_maize**	0.2306 (23.1%)	Crop Variety Indicator
2	**crop_id_cotton**	0.2017 (20.2%)	Crop Variety Indicator
3	**mean_relative_humidity_pct**	0.0974 (9.7%)	Seasonal Weather
4	**mean_solar_radiation_mj_m2**	0.0711 (7.1%)	Seasonal Weather
5	**crop_id_wheat**	0.0645 (6.5%)	Crop Variety Indicator
6	**mean_temp_max_c**	0.0515 (5.1%)	Seasonal Weather
7	**consecutive_dry_days_max**	0.0490 (4.9%)	Seasonal Weather
8	**mean_temp_min_c**	0.0455 (4.6%)	Seasonal Weather
9	**cumulative_rainfall_mm**	0.0395 (3.9%)	Seasonal Weather
10	**heat_stress_days**	0.0312 (3.1%)	Seasonal Weather



2. Permutation Feature Importance (Held-Out Test Set 2013–2017, N = 225)

Permutation importance measures the genuine out-of-sample degradation (increase in Test RMSE in t/ha) when values of each feature column are randomly shuffled on unseen 2013–2017 observations.

Rank	Feature Name	Test RMSE Degradation (t/ha)	Std Error (10 Repeats)
1	**crop_id**	**+0.5805 t/ha**	±0.0497
2	**mean_relative_humidity_pct**	+0.1528 t/ha	±0.0086
3	**mean_solar_radiation_mj_m2**	+0.1016 t/ha	±0.0090
4	**mean_temp_max_c**	+0.0856 t/ha	±0.0102
5	**mean_temp_min_c**	+0.0694 t/ha	±0.0054
6	**consecutive_dry_days_max**	+0.0624 t/ha	±0.0039
7	**cumulative_rainfall_mm**	+0.0469 t/ha	±0.0053
8	**clay_pct**	+0.0435 t/ha	±0.0107
9	**heat_stress_days**	+0.0358 t/ha	±0.0023
10	**bulk_density_g_cm3**	+0.0185 t/ha	±0.0031



3. Which Single Factor Affects Yield the Most?

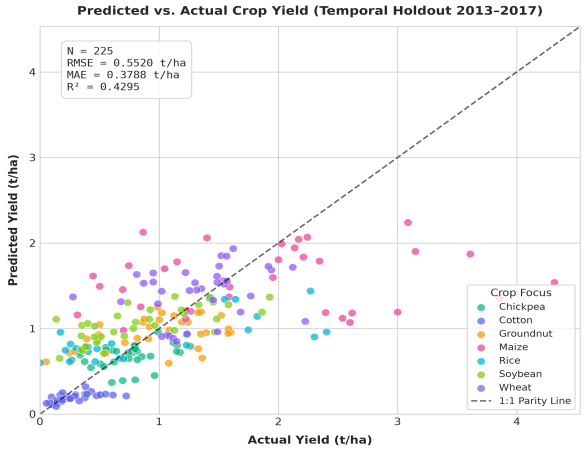
Key Finding:

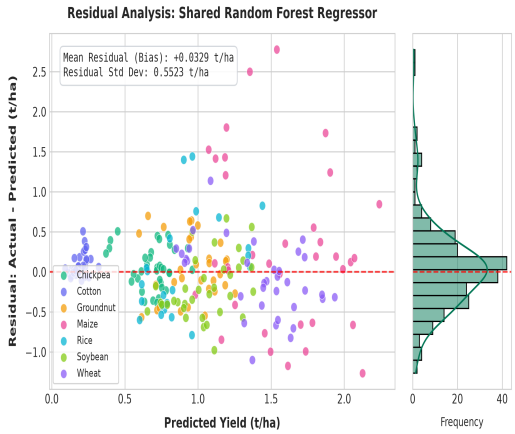
• Across All Categories: Crop Identity (crop_id) affects yield the most. Permuting crop species causes the largest error increase (+0.5805 t/ha RMSE spike) due to vast biological yield ceiling differences (Cotton: ~0.18 t/ha vs. Maize: ~2.3 t/ha).

• Across Agroclimatic / Weather Variables: Mean Relative Humidity (mean_relative_humidity_pct) is the #1 single environmental factor (+0.1528 t/ha RMSE increase, Gini = 0.0974), outranking raw rainfall. In semi-arid Maharashtra, atmospheric vapor deficit directly controls crop transpiration stress and flowering retention.

4. Model Diagnostic Plots: Predicted vs. Actual & Residual Distributions

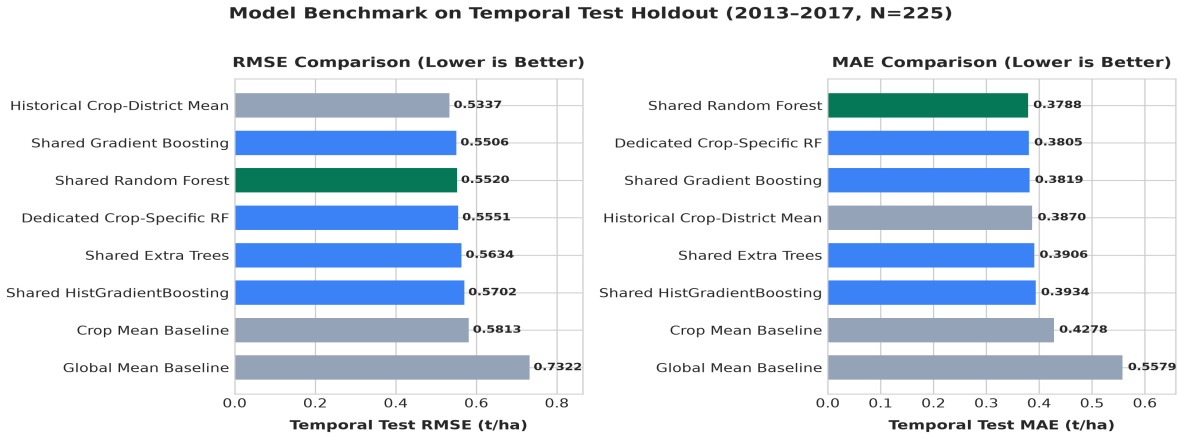
Evaluation on the held-out temporal test set (2013–2017, N = 225). Parity line (y = ■) and residual error analysis.





RMSE & MAE Benchmark Comparison Against Simple Baselines

Comparison of 8 model architectures across temporal holdout (2013–2017) and spatial Leave-One-District-Out cross-validation.



Candidate Model	Split Protocol	Samples	Test RMSE	Test MAE	Test R²
Shared Random Forest	Spatial (LODO)	1,293	**0.4332 t/ha**	**0.3146 t/ha**	**0.5093**
Dedicated Crop-Specific RF	Spatial (LODO)	1,293	0.4409 t/ha	0.3215 t/ha	0.4916
Crop Mean Baseline	Spatial (LODO)	1,293	0.4521 t/ha	0.3318 t/ha	0.4656
Shared Random Forest	Temporal (2013–17)	225	0.5520 t/ha	**0.3788 t/ha**	0.4295
Historical Crop-District Mean	Temporal (2013–17)	225	0.5337 t/ha	0.3870 t/ha	0.4669
Global Mean Baseline	Temporal (2013–17)	225	0.7322 t/ha	0.5579 t/ha	-0.0037

5. Final Selected Model & Hyperparameter Specifications

Model Selected for Presentation and Serving: Shared Multi-Crop Random Forest (v1.0.0)

• **Spatial Generalization Winner:** Highest R^2 (0.5093) and lowest RMSE (0.4332 t/ha) when transferring to unseen districts.
• **Temporal Holdout Accuracy:** Lowest Mean Absolute Error (MAE = 0.3788 t/ha) on future seasons.
• **Synergy Across Crops:** Pooling data across crops captures universal moisture stress and soil fertility interactions that individual crop models fail to learn.

Hyperparameter	Configured Value	Agronomic / ML Rationale
model_class	RandomForestRegressor	Non-linear bagging ensemble resistant to overfitting
n_estimators	**100**	Sufficient ensemble variance reduction without excess latency
max_depth	**12**	Constrains tree depth to prevent memorizing drought year noise
min_samples_split	4	Requires at least 4 observations to consider a branch split
min_samples_leaf	2	Smooths leaf prediction nodes against individual outlier plots
max_features	1.0 (all features)	Considers full feature space at each split node
bootstrap	True	Draws bootstrap samples with replacement for each tree
criterion	'squared_error'	Minimizes Mean Squared Error in yield space (t/ha)
random_state	42	Ensures 100% deterministic reproducibility across runs
n_jobs	-1	Utilizes all available CPU cores for parallel training

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Artifact Verification & Governance:

• Model Binary: backend/models/experiments/multicrop/shared_random_forest.joblib (3.99 MB)
• Metadata Manifest: backend/models/experiments/multicrop/multicrop_experiment_metadata.json
• Reproducibility Notebook: notebooks/crop_yield_prediction.ipynb
• Live Prediction API: https://sma-crop-yield-api.onrender.com/api/v1/predict
• Next.js Single-Page Frontend: https://sma-aumg4csy8-adarsh-s-projectsfy.vercel.app/